香港中文大學（深圳）
The Chinese University of Hong Kong, Shenzhen

Introduction to Data Science

Lecture 16 Optimization II: Convex Function

Outline

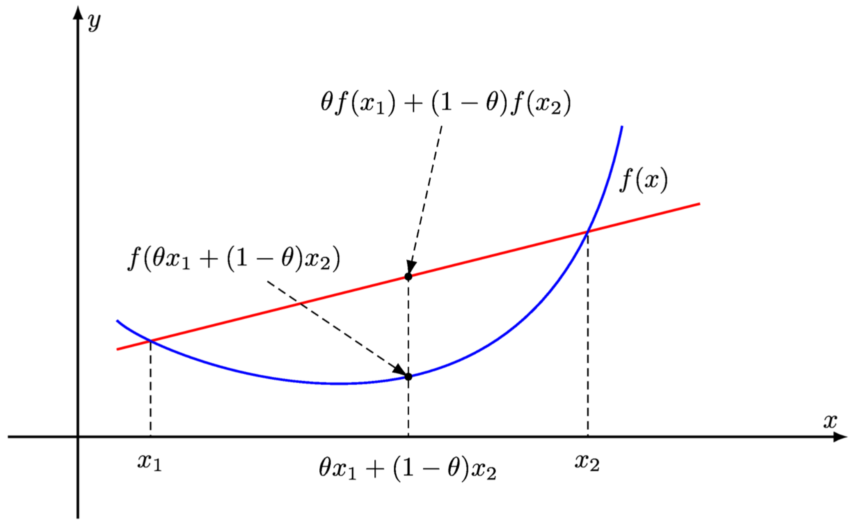
- Convex function
  - Definition
  - Geometry
  - Second-order Condition
- A Classic Optimization Problem: Newsvendor
  - Problem formulation

Convex Function: Definition

- A function $f(\boldsymbol{x}): R^n \to R \cup \{+\infty\}$ is convex if
    1) Its *domain* (the region where f is finite-valued), denoted as $dom(f)$, is a convex set
    2) For any $\boldsymbol{x_1}, \boldsymbol{x_2} \in dom(f)$ and any $0 \leq \lambda \leq 1$, we have

       $$f(\boldsymbol{z}) \leq \lambda f(\boldsymbol{x_1}) + (1-\lambda) f(\boldsymbol{x_2}),$$

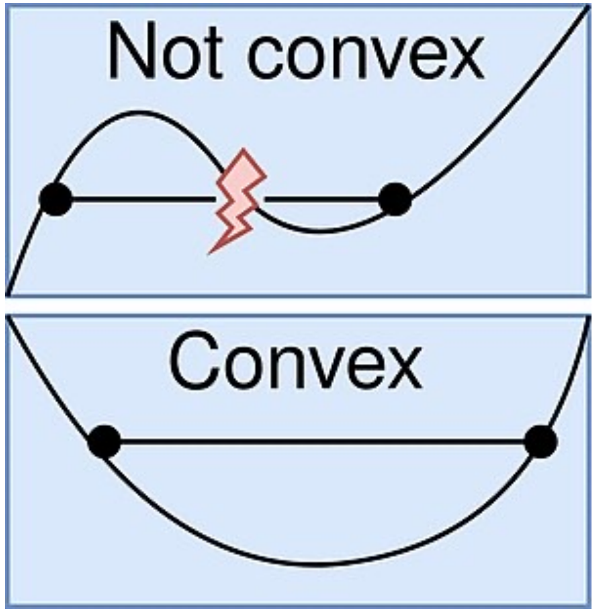
       where $\boldsymbol{z} = \lambda \boldsymbol{x_1} + (1-\lambda)\boldsymbol{x_2}$.



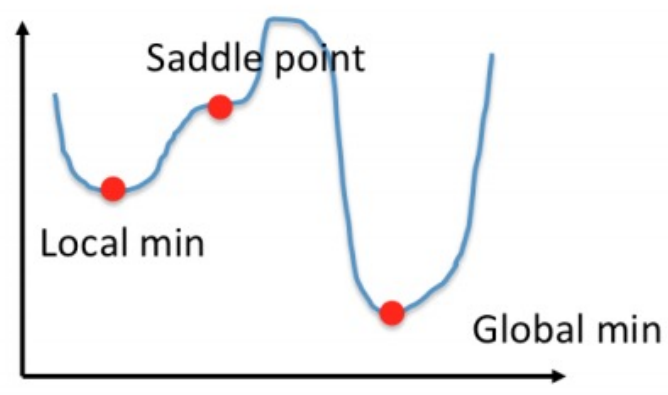
Convex Function: Geometric Interpretation

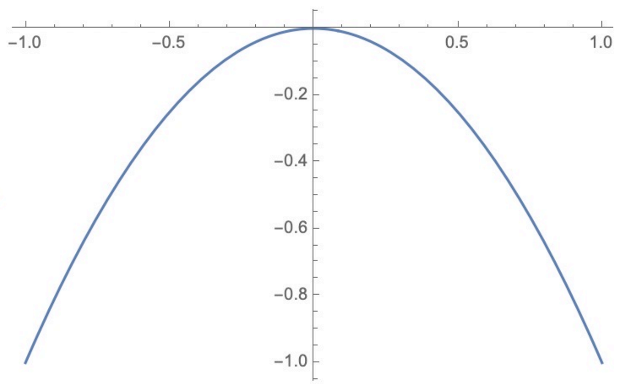
- The line segment connecting any two points on the function graph lies above the function curve.

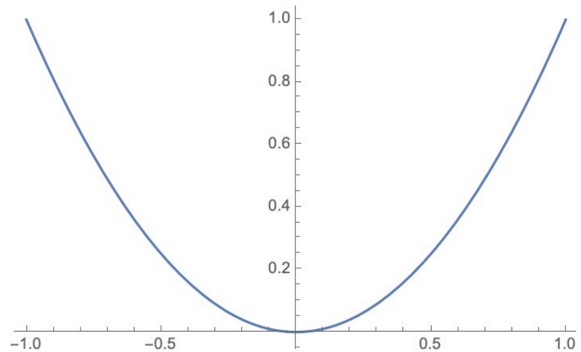
- The region above a convex function is a convex set.



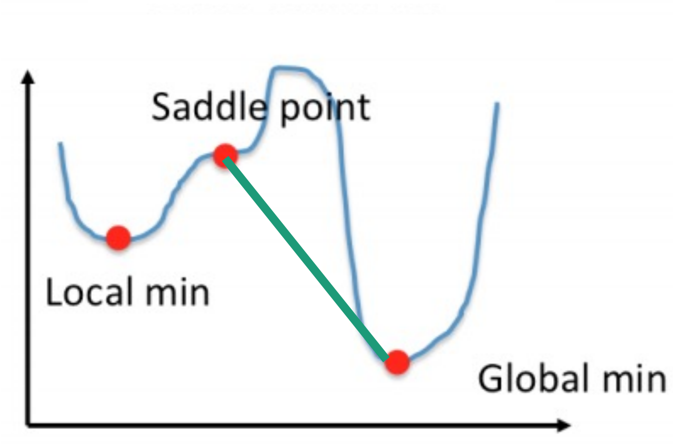
Exercise: Convex function?

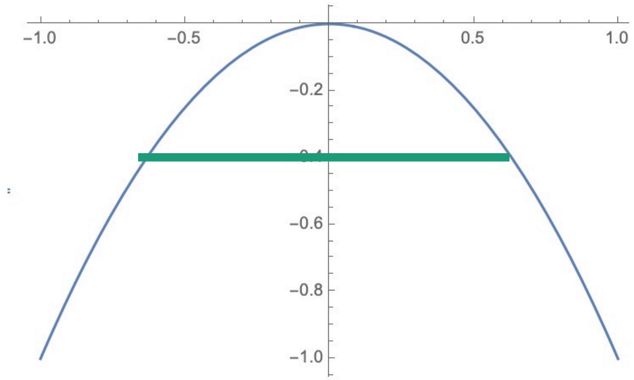


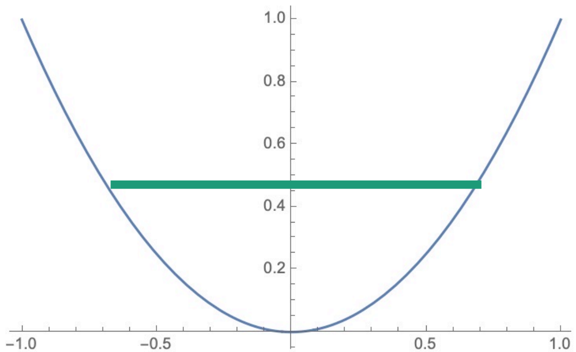




Exercise: Convex function?



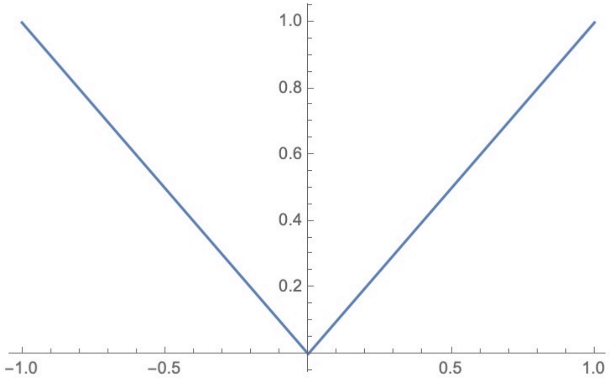




Exercise

- Prove $f(x) = |x|$ is convex by definition.

Exercise



- Prove $f(x) = |x|$ is convex by definition.

- Proof:
  - The domain of f is the entire real line, and thus is convex
  - Pick any two points x_1, x_2 and $\lambda \in (0,1)$, we have from triangle inequality
$$|\lambda x_1 + (1-\lambda)x_2| \leq |\lambda x_1| + |(1-\lambda)x_2| = \lambda|x_1| + (1-\lambda)|x_2|.$$

  This shows the convexity.

Proving Convexity using Second Order Condition

- (Univariate) Suppose $f: R \to R$ is twice continuously differentiable. Then f is convex if and only if $f''(x) \geq 0$.

Exercise

- Prove the following functions are convex
  - $f(x) = ax + b$
  - $f(x) = x^2$
  - $f(x) = e^x$

Exercise

- Prove the following functions are convex
  - $f(x) = ax + b$
    - $f''(x) = 0$
  - $f(x) = x^2$
    - $f''(x) = 2$
  - $f(x) = e^x$
    - $f''(x) = e^x > 0 \text{ for all } x$

Proving Convexity using Second Order Condition

- (Multivariate) Suppose $f: R^n \to R$ is twice continuously differentiable. Then f is convex if and only if for any $\boldsymbol{x}$, for every direction $\boldsymbol{e}$, it holds that

$$\left.\frac{d^2 f(\boldsymbol{x} + t\boldsymbol{e})}{dt^2}\right|_{t=0} \geq 0$$

  - This means that along every direction $\boldsymbol{e}$, the univariate function $t \mapsto f(\boldsymbol{x} + t\boldsymbol{e})$ is convex

Example

- Show that $f(x_1, x_2) = x_1^2 + 3x_2^2$ is convex.

- Proof.
  - Pick any direction (e_1, e_2). Consider a function
$$F(t) = f(x_1 + te_1, x_2 + te_2) = (x_1 + te_1)^2 + 3(x_2 + te_2)^2.$$
  - Then we have
$$F'(t) = 2te_1^2 + 2x_1e_1 + 6te_2^2 + 6x_2e_2$$
$$F''(t) = 2e_1^2 + 6e_2^2 \geq 0$$

  Hence $F(t)$ is convex in t.
  - This holds for every direction e, and thus f is convex.

Operations that Preserve Convexity

- If f is convex, then f/c is convex for any $c > 0$

- If f_1, f_2 are convex, then $f_1 + f_2$ are convex

Concave Function: Definition

- A function $f(\boldsymbol{x}): R^n \rightarrow R \cup \{+\infty\}$ is concave if
  1) Its *domain* $dom(f)$, the region where f is finite-valued, is a convex set
  2) For any $\boldsymbol{x_1}, \boldsymbol{x_2} \in dom(f)$ and any $0 \leq \lambda \leq 1$, we have
  $$f(\boldsymbol{z}) \geq \lambda f(\boldsymbol{x_1}) + (1-\lambda) f(\boldsymbol{x_2}),$$
  where $\boldsymbol{z} = \lambda \boldsymbol{x_1} + (1-\lambda)\boldsymbol{x_2}$.

- f is concave, if and only if $-f$ is convex

Exercise

- Prove the following functions are concave:
  - $f(x) = -x^2$
  - $f(x) = \log(x)$ on $(0, +\infty)$
  - $f(x) = \sin(x)$ on $[0, \pi]$

Exercise

- Prove the following functions are concave:
  - $f(x) = -x^2$
    - $f''(x) = -2 < 0$
  - $f(x) = \log x$
    - $f''(x) = -\frac{1}{x^2} < 0 \text{ on } (0, +\infty)$
  - $f(x) = \sin(x)$ on $[0, \pi]$
    - $f''(x) = -\sin(x) \leq 0,\ x \in [0, \pi]$

Example: The Mysterious Bakery: A Story of Demand, Orders, and Profit 🍞

- Imagine you own a small bakery in town, famous for its delicious croissants. Every morning, you have to decide how many croissants to bake. Too many, and you might waste food. Too few, and you'll miss out on potential sales!

- Now, let's turn this real-life situation into an optimization problem

Step 1: What Are We Deciding? (The Decision Variable 🎯)

Your main decision each morning is:

How many croissants should I bake?

Let's call this number Q.

- If you bake too many, some croissants might go unsold and get **stale**.
- If you bake too few, you might have **disappointed customers** and lost revenue.

So, we need to **find the perfect** Q that maximizes our bakery's profit!

Step 2: The Mysterious Demand (Randomness in the Real World 🎲)

- But here’s the challenge: **you don’t know exactly how many customers will come each day**!
  - Some days, the town is **buzzing with tourists**, and you sell out quickly.
  - Other days, it’s **rainy and quiet**, and only a few locals stop by.

Let’s call this **demand** D.

- Since we **don’t know D in advance**, we treat it as a **random variable**. That means we can only make **an educated guess** about how many croissants people will buy.

Step 3: How Do We Make Money? (Profit Calculation 💰)

Let's say:

- p = The price you sell **each croissant** for ($2 per croissant).
- c = The cost to bake **each croissant** ($0.50 per croissant).
- s = The discount price if you have **leftover croissants** that are sold the next day at a discount ($1 per croissant).

- Now, two things can happen:

Case 1: You Sell Out (D ≥ Q) 🚀

- If demand is higher than or equal to your baked quantity $(D \geq Q)$, you sell everything!
- Your profit is:

$$\text{Profit}=(p-c)Q$$

(Since every croissant is sold at full price.)

Case 2: You Have Leftovers (D < Q) 😞

- If demand is lower than your baked quantity $(D < Q)$, you **only sell** D **croissants** at full price.
- The rest $Q - D$ are sold at the **discount price** s.
- Your profit is:

$$\text{Profit} = (p - c)D + (s - c)(Q - D)$$

Step 4: Writing the Profit Formula (Mathematics Behind the Bakery 🍩📜)

- To simplify, we define:

$$(Q-D)^{+}=\max(Q-D,0)$$

which represents the number of **unsold croissants**.

- Now, your profit formula looks like:

$$\text{Profit} = p\min(Q,D)+s(Q-D)^{+}-cQ$$

- Since demand D is random, we calculate the **expected profit**:

$$f(Q)=p\,E[\min(Q,D)]+s\,E[(Q-D)^{+}]-cQ$$

Now, our goal is **to find the best Q that maximizes expected profit**.

Step 5: The Optimization Problem (Finding the Best Order Quantity 🏆)

We want to:

$$\max_{Q} f(Q)$$

which is equivalent to:

$$\begin{aligned} &\text{minimize} \quad && -(pE[\min(Q, D)] + sE[(Q - D)^+] - cQ) \\ &\text{subject to} \quad && -Q \leq 0 \end{aligned}$$

(You can't bake a negative number of croissants!)

Step 6: Why This Is a Nice Problem (And Why You'll Get the Best Answer 🍀)

The Profit Function is Concave → Convex Optimization 🎯

- The expectations involve simple linear functions, which makes the problem **convex**. Think about why.
- Convex problems are great because they always have a **unique global solution**! 🎉

The Feasible Region is Convex (No Weird Solutions) 🏠

- The constraint $Q \geq 0$ defines a **half-space**, which is convex.
- That means we don't have to worry about strange or "bad" solutions—**there's only one best answer**.

Step 7: Solving the Problem (How to Find the Best Q? 🔍)

1.Use Calculus

- If the demand D follows a known distribution (e.g., **normal** or **Poisson**), we can compute expectations and set the derivative of $f(Q)$ to **zero** to find the best Q.

2.Use a Solver (Computers to the Rescue! 🤖)

- If the demand is complicated, we can use numerical solvers like **SciPy (Python)**, **Excel Solver**, or **Gurobi** to find the best Q.

Hint: How to show the objective function is convex?

minimize $\boxed{-(pE[\min(Q,D)] + sE[(Q-D)^+] - cQ)}$

subject to $-Q \le 0$

$\min(Q,D) = \mathrm{Q} - (\mathrm{Q} - \mathrm{D})^+$

Do not require differentiability of the cdf of D

$$\boxed{\overbrace{(p-c)(-Q) + (p-s)E[\underbrace{\max(Q-D,0)}_{\text{maximum of convex functions (in Q)}}]}^{\text{nonnegative weighted sums of convex functions (in Q)}}}$$

Convex!

Final Thoughts: Why This is Cool 🎉

By solving this problem, you can:
✅ **Predict how much to order in uncertain situations** (like bakeries, online stores, or even airlines!).
✅ **Understand how demand randomness affects profits**.
✅ **Use math to make real-world decisions smarter!**

Would you like a Python code to simulate this and find the best Q? 🚀

Summary

- Prove a function is convex
  - By definition
  - By geometry
  - By second-order condition